SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: MLA 02 COURSE NAME: Fundamentals Of Machine Learning

COURSE OUTCOME COVERED

CO5: *Create graphical models to analyze and infer patterns in complex probabilistic systems for structured decision-making.(BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 10 Total Marks:50 Annexure A

Mock Interview

Code of Conduct:

*I **G Chandravardhan Reddy - 192425037**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: *chandra*

Annexure A

Mock Interview

Q1. “Suppose you are working on an industrial monitoring system. How would you use Bayesian Networks to predict machine failures?” (5 Marks – 8 Mins)

Q2. “Imagine you are asked in an interview to design an image denoising solution. How would Markov Random Fields help model neighborhood pixel dependencies?” (5 Marks – 8 Mins)

Q3. “If a company wants to build a speech recognition system, how would you apply a Hidden Markov Model to process sequential audio signals?” (5 Marks – 8 Mins)

Q4. “During a technical interview, how would you explain the use of Conditional Random Fields for sequence labeling tasks such as part-of-speech tagging?” (5 Marks – 8 Mins)

Q5. “Suppose an interviewer asks how conditional independence simplifies probabilistic inference. How would you justify it with a graphical model example?” (5 Marks – 8 Mins)

Q6. “In a healthcare diagnosis problem, how would you compare directed graphical models and undirected graphical models for representing symptom relationships?” (5 Marks – 8 Mins)

Q7. “If patient symptom data is partially missing, how would you perform Bayesian inference to still make an accurate diagnosis?” (5 Marks – 8 Mins)

Q8. “How would you use a Hidden Markov Model to analyze customer purchase sequences in an e-commerce recommendation system?” (5 Marks – 8 Mins)

Q9. “In an NLP interview task, how would you explain why Conditional Random Fields often outperform Hidden Markov Models in named entity recognition?” (5 Marks – 8 Mins)

Q10. “Suppose you are asked to improve fraud detection. How would graphical model learning and generalization enhance prediction accuracy?” (5 Marks – 8 Mins)

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively

Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

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CO5-AT3

1. Suppose you are working on an industrial monitoring System. How would you are use Bayesian Networks to predict machine failure?

A Bayesian networks represent relationship between machine conditions like temparature, vibration and pressure. Using Sensor data and probabilites they Calculate the Probability of machine failure. This helps in early warning and preventive maintanance.

2. Imagine you are asked in an interview to design an image denoising solution. How would Markov Random field helps model neighbourhood pixel dependencies?

A MRF model the relationship between a pixel and its neighbouring pixels. They assume nearby pixels usually have similar values. By monitoring and minimizing an early energy function, noise can be removed while preserving the image.

3 If a Company wants to build a speech recognition System, how would you apply a Hidden Markov Model to process Sequential audio signals?

A HMM represents speech sounds as hidden states and audio features as observed observations. It uses transition and emission probabilities to find the most likely sequence of sounds using the Viterbi algorithm.

4 During a technical interview, how would you explain the use of Conditional Random field for Sequence labeling tasks such as speech tagging?

A CRF assigns labels such as noun, verb & adjective to words. Its uses word features and relationships between neighbouring labels to find the most likely sequence of Pos tags.